

incorporated into a binary arrangement of daemons.

Some core components of SaltStack master are:

- **SaltMine:** SaltMine gathers the state information of the Salt minions and stores it in the master, and then makes the data accessible back to the minions. Mine data basically contains the network, firewall, and more.
- **SaltPillar:** This is used to put down delicate information like SSH key, credential files and passwords.
- **Auth service:** This service is used to authenticate the minions' communication.
- **File server:** This is the ZeroMQ server. We can have ZeroMQ exchange the files from Salt master to minions. File worker requires a backend.
- **Salt wheel (API):** Salt wheel permits developers to programmatically get to the Salt master.
- **Salt cloud:** Salt cloud is a powerful interface for cloud hosts. Thus, it will give you the communication between Salt master and cloud machines.
- **Salt runner:** Salt runners are modules that assist with executing orders from the Salt master.
- **Salt reactor:** Salt reactor is an interface that keeps track of the Salt event bus for the particular event tag.
- **Salt engine:** This is the external system that uses the capacity of Salt expert's long-running process. It controls Salt configurations, execution modules and runners. Engines are constantly associated with the event bus.
- **Salt SSH:** Salt SSH is an in-built service that permits the Salt master access to the Salt minions.
- **Grain:** Salt grain is an interface, which resembles a Salt mine. However, it will have static data of minions like operating system, domain name, IP address bit, OS type, memory, disk space, and numerous other system properties. In contrast to the Salt mine, you can build your own custom Salt grain to have the particular data about the Salt minion.
- **Returner:** Return data is the chunk of information that is returned after the execution of the command sent from the Salt master to Salt minions. Returner is an interface that permits the return information to be shipped off to any framework like the logical framework or monitoring system.
- **Salt beacon:** Salt beacon is the surveillance system of the Salt minions. These beacons are completely developed monitor and alert frameworks that will catch the issue or event, and permit Salt reactor to trigger accordingly.
- **Salt execution module:** When the Salt master gives the commands, execution modules from the Salt minions guarantee the execution and constant monitoring of these commands.

Installing SaltStack

Prior to installation, you need to have the following prerequisites:

- Linux server
 - Root access to the server
 - The SaltStack repository
- Install all the updates by using the command given below:

```
sudo apt-get update
```

Install SaltMaster from the repository using the *apt-get* command:

```
sudo apt-get install salt-master
```

Install the Salt minion from the repository by using the following *apt-get* command:

```
sudo apt-get install salt-minion
```

Install the *salt syndic* from the repository with the following command:

```
sudo apt-get install salt-syndic
```

Master configuration

Salt configuration is not at all complicated. The configuration files are already installed in the */etc/salt* directory and are named after their respective parts — for instance, */etc/salt/master* and */etc/salt/minion*:

```
#interface: 0.0.0.0
interface: <local ip address>
```

After updating the configuration file, restart the Salt master by using the command:

```
sudo service salt-master restart
```

Minion configuration

The configuration of a Salt minion is very easy. A Salt minion attempts to connect to the DNS named *salt*; if the minion is able to sort out that name, no configuration is needed. Redefine the master directive in the minion *config* file, which is */etc/salt/minion*, as shown below:

```
#master: salt
master: <local ip address>
```

After updating the configuration file, restart the Salt minion by using the command:

```
sudo service salt-minion restart
```

Key management

Salt uses AES encryption for all the interactions between the master and the minion. The communication between the master and minion is authenticated and keys are accepted.

```
salt-key -L
```

The output for the above command

Components of SaltStack minions

Salt minions have the following chief components that make the configuration management system complete.